YUFU

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EDUCATION BACKGROUND

Sichuan University, Pittsburgh Institute *Sep 2023 – Present*

Bachelor of Engineering in Computer Science from Sichuan University June 2027

Overall GPA: 90.26/100, 3.84/4

HONORS AND AWARDS

First-Class Scholarship, Sichuan University (Top 5% in major, academic & moral excellence) *2025*

College Students' Innovative Entrepreneurial Training Plan Program, Province Prize *2025*

National Scholarship, Sichuan University (Top 2% in major) *2024*

Dean's List, Sichuan University (Top students in Pittsburgh Institute) *2024*

PUBLICATIONS

- **Yu Fu**, Linyue Cai, Ruoyu Wu, Yongqi Kang, Yong Zhao*, *"Nutrient-Aware Personalized Meal Recommendation Using Structured Food Knowledge and Constraint Verification"*, Foods, 15(10), 1647, DOI: 0.3390/foods15101647
- Linyue Cai, Chaojia Yu, Yongqi Kang, **Yu Fu**, Heng Zhang, Yong Zhao*(2025), "*Practices, opportunities and challenges in the fusion of knowledge graphs and large language model*s",Frontiers in Computer Science, 7, 1590632, DOI: 10.3389/fcomp.2025.1590632
- **Yu Fu**, Yongqi Kang,Yong Zhao*, "*Is AI Smart Enough? An Overview Regarding How Large Language Models Think and Reason*", NIP@IR Workshop (SIGIR 2025), Accepted, No proceedings.
- Huiming Wang, Leilei Ouyang, Chang Xia, Yongqi Kang, **Yu Fu,** Yuqi Ouyang*, *"AllocMV: Optimal Resource Allocation for Music Video Generation via Structured Persistent State"*, F2S@ICML 2026, Accepted, https://arxiv.org/abs/2605.10723.
- Linyue Cai, Yongqi Kang, Chaojia Yu, **Yu Fu,** Heng Zhang, Yong Zhao*, "*Bringing Two Worlds Together: The Convergence of Large Language Models and Knowledge Graphs*", International Conference on Automation, Robotics and Computer Engineering (ICARCE) 2024, DOI: 10.1109/ICARCE63054.2024.00046.

RESEARCH EXPERIENCE

Health-Promoting Kitchen Assistant for Aging Population *Jan.2026 – Present*

- Built the core recommendation engine integrating food knowledge graphs with medical dietary guidelines, enabling constraint-aware, health-oriented recipe suggestions.
- Engineered the full-stack system architecture (backend APIs, frontend interface) with a focus on accessibility, laying the groundwork for future multimodal interaction capabilities.
- Designing vision and voice interaction pipelines: planning to integrate computer vision for ingredient detection and large language models for voice-based conversational guidance, enabling natural, hands-free kitchen assistance.
- Incorporating emotional companionship features through an empathetic dialogue agent to improve user adherence and quality of life.

EduRiskX — Interpretable Student Risk Prediction Framework *Oct.2025-Feb.2026*

- Designed and implemented EduRiskX, an interpretable student risk prediction framework integrating Transformer-based time-series modeling with rule-supported explanations.
- Achieved early detection performance on the OULAD dataset, identifying at-risk students up to 19 weeks in advance, substantially earlier than strong deep baselines such as PatchTST and iTransformer (typically 15–16 weeks or later)
- Maintained a high value of AUC, F1-Score, precision and recall while providing theory-aligned explanations and actionable intervention strategies to enhance the practical effectiveness of early warning systems in education

West China Hospital Biomedical Big Data Center | Research Assistant Intern *Jun. 2025-Nov. 2025*

- Collaborated with the Department of Mental Health to develop an intelligent auxiliary diagnosis system integrating knowledge graphs and multimodal signal analysis for mental health assessment.
- Constructed a domain-specific knowledge graph via entity/relation extraction; engineered a multimodal emotion recognition module fusing speech emotion features and physiological signals to enhance analytical robustness.

- Validated system efficacy through clinical deployment at West China Hospital, supporting real-world mental health interventions.

LLM Reasoning Mechanism Optimization *Jan. 2025-Dec. 2025*

- Surveyed and critiqued state-of-the-art LLM reasoning techniques (CoT, ToT), identifying key challenges in their reliability for complex task.
- Proposed a novel theoretical model integrating Bayesian probability with LLM reasoning to enable more transparent and quantifiable assessments in educational settings.

Multimodal Mental Health Interaction System Development *Oct. 2024–Dec. 2025*

- Designed and implemented a deep learning model integrating speech and text for high-precision emotion recognition.
- Integrated modules into an end-to-end platform, achieving performance significantly superior to unimodal baselines.

Intelligent Recipe Recommendation System *May. 2025–Sep. 2025*

- Led the design and development of a knowledge graph + RAG system for personalized meal recommendation, handling fuzzy queries with nutritional awareness.
- Integrated USDA nutrient data with Recipe1M+; used a 1.5B LLM for intent parsing and an agentic critic for safety verification (98.5% constraint satisfaction).
- Achieved 0.825 semantic recall@5 on complex fuzzy intents, surpassing dense retrieval by 50%.

Sarcopenia Risk Prediction Model *Mar. 2024–Jan. 2025*

- Built a machine learning model to predict sarcopenia risk based on a Western Chinese cohort database.
- Cleaned clinical data and constructed a multi-dimensional feature system from thousands of records.
- Optimized models (SVM, Random Forest, XGBoost) and validated generalization across populations.

INTERNSHIP EXPERIENCE

Institute of Human-Computer Interaction and Media Integration, Tsinghua University | Algorithm Intern *Mar.2026– Present*

- Developed core functionalities for an enterprise-grade digital customer service system:

WeChat-Feishu Data Integration: Enabled automated replies in WeChat groups and synchronized all conversation logs (user, AI, human expert) to Feishu sheets, providing a foundation for service quality monitoring and data-driven optimization.

Human-like Delayed Response: Designed and deployed a delay mechanism that mimics natural typing/thinking pace.

- Adaptable to various enterprise scenarios; further customization for specific domains (psychological support, after-sales service) is underway.

Ernest & Young | AI & Data Intern *Jan.2026 - Mar.2026*

- Conducted research and requirement analysis for new features of EY's in-house AI platform EYC.ai, and implemented an invoice recognition function, covering the full pipeline from business requirement decomposition to feature delivery
- Contributed to the Text-to-Slides generation module of EYC.ai from 0 to 1, including product research, reproduction and evaluation of open-source solutions, and front-end rendering and system construction tailored to company requirement

Qi An Xin Technology Group Inc. | Testing Intern *Jul. 2025 - Sep. 2025*

- Conducted security vulnerability remediation and testing environment maintenance for key clients (e.g., Beijing TV Station).
- Independently performed patch installation and environment configuration; optimized internal security testing processes to improve coverage and efficiency.

EXTRACURRICULAR ACTIVITIES

Robot Soccer Club | R&D Director *Sep. 2024 - Jul. 2025*

- Conducted C programming and core algorithm training for new members.
- Led strategy design and code implementation for robot soccer matches; organized university-level AI and robotics competitions.

West China Association of Smart Stomatology| Deputy Director of the Organization Department *Sep. 2024 - Jul. 2025*

- Organized association events and shared frontier research insights in smart healthcare to keep members informed and connected.

PROFESSIONAL SKILLS

- Skills: Java, C / C++, Python, Go, SQL; SPSS, MATLAB
- Tools: Latex, Excel, PPT,Word, Xmind
- Languages: Chinese Mandarin (Native), English (IELTS: 7.0), Spanish(Basic,A2)